Teacher Wu Kuan | See It at a Glance #23 — The Circle Equation Is Just Pythagoras

English lecture notes · YouTube Short aRtrJiud38c

1. The trap question

Why is a circle $(x - h)^2 + (y - k)^2 = r^2$?

2. The method

- Every point on the circle is distance r from the center (h, k).
- Horizontal gap $x - h$ and vertical gap $y - k$ are legs; r is the hypotenuse.
- Read the center by flipping signs: $(x - 2)^2 + (y + 3)^2 = 25 \rightarrow$ center $(2, -3)$, radius 5.
- Tangent line ⇔ distance from center to line = radius.

3. Parent check method

Ask: “For $(x + 1)^2$, what is h?” (−1 — the sign flips.)

Practice

1. Find the center and radius of $(x + 1)^2 + (y - 4)^2 = 9$.
2. Find the center and radius of $x^2 + y^2 - 4x + 6y - 12 = 0$.
3. Write the circle with center (1, 2) passing through (4, 6).
4. Is (5, 2) inside, on, or outside $(x - 2)^2 + (y + 3)^2 = 25$?
5. The line $3x + 4y = k$ is tangent to $(x - 2)^2 + (y + 3)^2 = 25$. Find all k.

Answers

1. Center $(-1, 4)$, radius 3
2. Center $(2, -3)$, radius 5
3. $(x - 1)^2 + (y - 2)^2 = 25$
4. Outside ($9 + 25 = 34 > 25$)
5. $k = 19$ or $k = -31$

Saber Math · Teacher Wu Kuan (YouTube: Saber Math) · Free first-week trial of Kuan Weekly practice (email sign-up): https://kuan-weekly.vercel.app/sample · More: https://kuanmath.vercel.app · SAT notes: DM IG @saber_math